

Core: Requirements

CHAPTER 3: FINALIZING THE REQUIREMENTS

Chapter Objectives

In this chapter, we will prepare requirements for allocation to a release by:

- Prioritizing requirements using a sensible methodology
- Organizing requirements to create logical groupings
- Identifying ways to document requirements

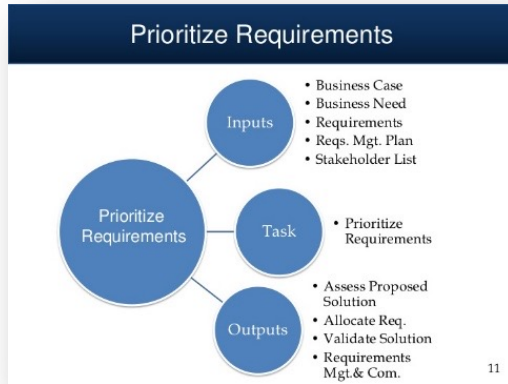
Chapter Concepts

Prioritizing Requirements

Organizing Requirements

Documenting Requirements

Prioritizing Requirements



- ◆ Prioritization of requirements ensures that analysis and implementation efforts focus on the most critical requirements
 - Critical to the Agile Scrum approach
- ◆ Prioritization is initially based upon key elements:
 - Business value
 - Business risk
 - Technical risk
 - Implementation feasibility
 - Operational feasibility
 - Time constraints
 - Budget constraints
 - Regulatory compliance
 - Policy compliance
- ◆ Prioritization methods are often project dependent

Prioritization Challenges

- ◆ An analyst faces several prioritization challenges
- ◆ Non Negotiable Demands
 - Stakeholders avoid difficult choices, fail to recognize the necessity for making tradeoffs, or attempt to rank all requirements as high priority
- ◆ Unrealistic Tradeoffs
 - Prioritization process influenced by unintentional over-estimation of the difficulty or complexity of implementing certain requirements
- ◆ Decision (or request) to deliver tactical vs. strategic solutions



Strategic vs. Tactical Solutions

Strategy	Tactics
Planning	Doing
Large Scale	Smaller Scale
Why	How
Difficult to Copy	Easy to Copy
Long Time Frame	Short Time Frame

◆ Strategic solutions

- Support the long-term aims and objectives of the business
- Expect costs to be spread over a long period of time
- Expect benefits to be spread over a long period of time
- Holds the potential for the biggest improvements
- Can cause the biggest disasters if not done correctly

◆ Tactical solutions

- Look at solving a specific problem the business is having
- Expect costs to be spread over a short period of time
- Expect benefits to be spread over a short period of time
- May solve a problem in the near term that has to be re-worked for a strategic solution

Two Common Prioritization Techniques

- ◆ High/Medium/Low (or 1/2/3)
 - **High:** a mission critical requirement; required for next release
 - **Medium:** supports necessary system operations; required eventually but could wait until a later release if necessary
 - **Low:** enhancement nice to have someday if resources permit
- ◆ Essential/Conditional/Optional
 - **Essential:** the product is not acceptable unless these requirements are satisfied
 - **Conditional:** would enhance the product, but the product is not unacceptable if absent
 - **Optional:** functions that may or may not be worthwhile



MoSCoW Prioritization Technique

- Divide your outcomes into four categories analyzed for business value
 - Driven by the customer
 - Relative to a given release cycle or relative to the solution scope itself

Must

- Outcome must be delivered in solution
- Critical to solution's success

Should

- Outcome included in the final solution if necessary

Could

- Outcome desired in the final solution but not critical

Won't

- Outcome not implemented in this release
 - May be considered in next phase

MoSCoW Best Practice

- MoSCoW analysis works well in a brainstorm discussion meeting
- Stakeholders should be physically present for priority setting meeting
- Walk through requirements and identify each one with an M, S, C, or W next to each (different color pen and flip chart helps)
- A way of helping focus participants is to time-box the target, for example by asking them to allocate the categories based on what must/should/could be done within the next four weeks
- An analyst can attempt an initial prioritization individually
 - Stakeholders then clarify and reprioritize
 - Prioritizing may lead to additional requirements, so even prioritization can be considered an elicitation technique!



Activity 3.1: MoSCoW

Goal	Determine the value and use of MoSCoW
Team Format	Class discussion
Materials	This slide
Tasks	1. Is MoSCoW a realistic prioritization technique or not? 2. A requirement document should only contain items which are required for the final system. Therefore, what is the point in having any categories besides Will or Won't? 3. Within the context of Agile development, can MoSCoW be used successfully?
Deliverable	Research and be prepared to comment on examples of MoSCoW and Agile being used together
Duration	15 minutes

Prioritization in Agile

- Often uses a technique called “Brutal Prioritization”
 - Keep backlog lean and properly prioritized
 - Not letting team members (customers) force their own personal requirements
 - Requirements whose technical feasibility is not determined
 - Weak features that may degrade the product
- MoSCoW used frequently
 - Team velocity can be used to help determine what can be accomplished during a sprint
 - Consider the use of a “spike solution” if difficult to estimate a story
- May also use the Kano model
 - Must-haves: Cannot ship product without it
 - Dissatisfiers: Something that must *not* be included
 - Satisfiers: Each one adds incremental value
 - Delighters: Boosts customer satisfaction and recommendation

Activity 3.2: Agile Prioritization Techniques



Goal	To explore a variety of techniques which may be used to prioritize requirements when using Agile
Team Format	Six teams
Materials	Laptops, flip charts
Tasks	1. Your team will be assigned one of the following techniques • Relative weighting • Theme screening • Theme scoring • Stacked ranking • Context evaluation • Value mapping 2. Research this topic area and prepare a short presentation (less than 5 mins) 3. Select a spokesperson to present your research
Deliverable	Flip chart of results
Duration	35 minutes

Activity 3.3: Additional Prioritization Methods



Goal	Describe additional requirement prioritization methods
Team Format	Five teams
Materials	This slide, the Internet
Tasks	1. The instructor will assign each team one of the following prioritization methods A. 100-Point Method B. Binary Search Tree C. Numeral Assignment Technique D. Theory-W E. Wiegers' Method 2. Research the method 3. Demonstrate the technique to the class
Deliverable	Description of additional prioritization methods
Duration	30 minutes

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Organizing Requirements

- ◆ Significant amounts of information will be discovered during the elicitation and modeling processes
- ◆ The analyst must spend time organizing requirements into a useable whole that illuminates priority and interrelationships
- ◆ Two key objectives when organizing requirements
 - Understand which models are appropriate for the business need
 - Identify interrelationships and dependencies
 - ◆ Singular requirements are rarely complex
 - ◆ Interdependencies between requirements add the element of complexity



Technique: Organize by Usage

Organizational

Requirements can be arranged around the needs of each stakeholder group

Process

Requirements can be organized around relevant end-to-end business processes

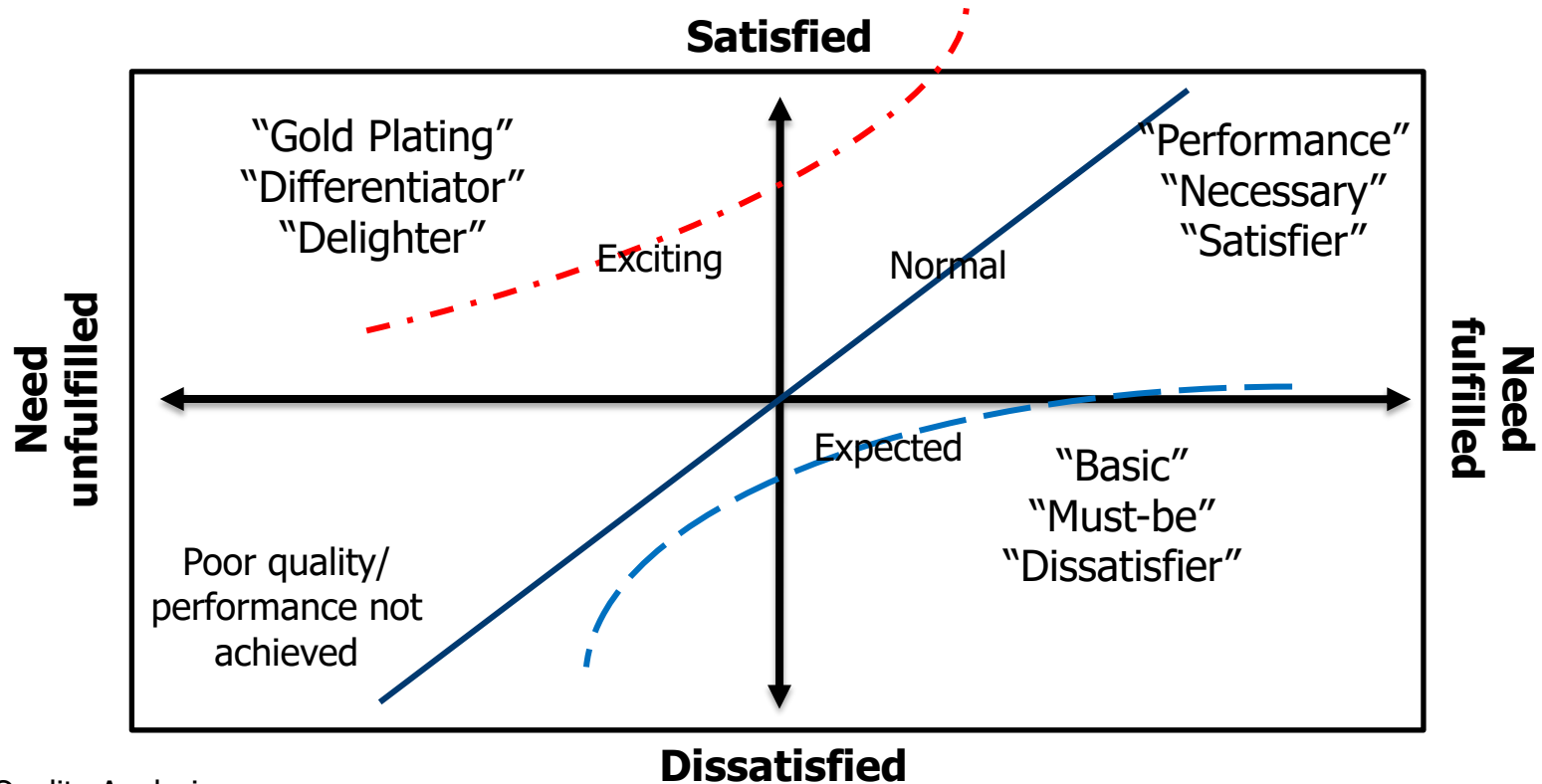
Scenarios

Requirements can be organized around the individual goals of how a stakeholder group will use the requirements

"As humans most of us cannot visualize the relationships between the many requirements (especially if the system is large) in our head. Organizing them and prioritizing them creates a structure showing the relationships between requirements, whether it is hierarchical, shows a relationship, or shows conflict."

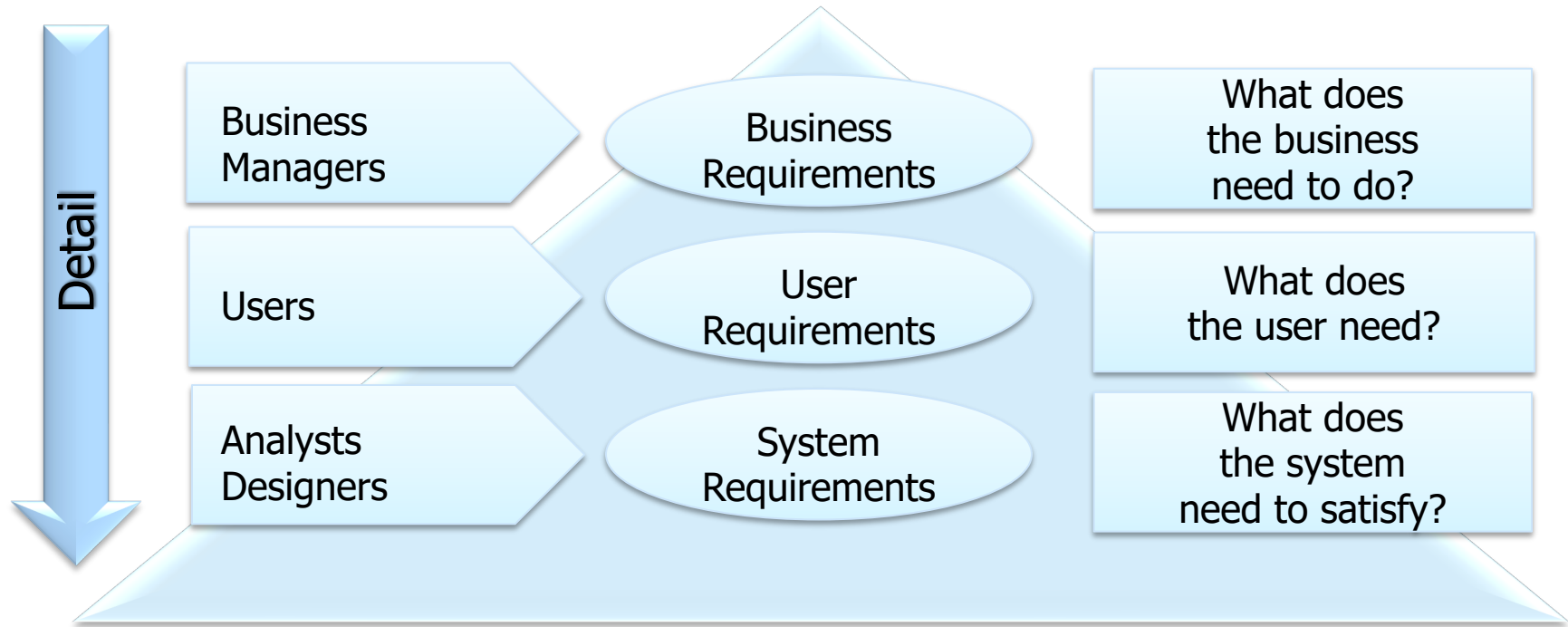
Chad La Fournie of the Software Engineering Research Network

Technique: Organize by Quality



*Kano Quality Analysis

Technique: Organize by Audience



Chapter Concepts

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Requirements Documentation

“The purpose of war is not battle but victory.”

Sun Tsu, The Art of War, ca 500 BC

... said another way ...

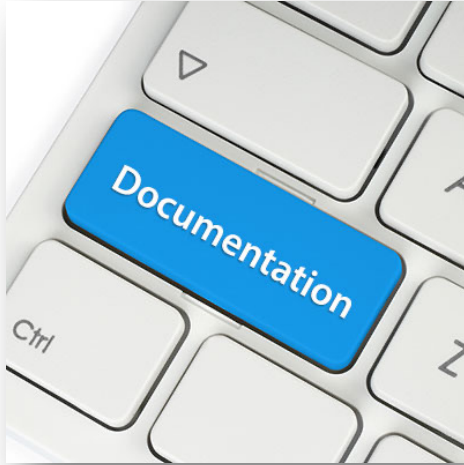
The purpose of documentation is not to prove the Analyst's
ability to produce elaborate tables and text...
the purpose is to facilitate understanding.

The Objectives of Documentation

- ◆ Requirements documentation is used to meet multiple objectives
 - Express stakeholder desires (requirements) via a combination of text-based statements, tables, prototypes, or models
 - Guide the development and implementation of solutions
 - Facilitate stakeholder communication in a format they will best understand
 - Support training activities and knowledge transfer
- ◆ Documentation can be generated at any point in the process
 - Chunks
 - Complete components



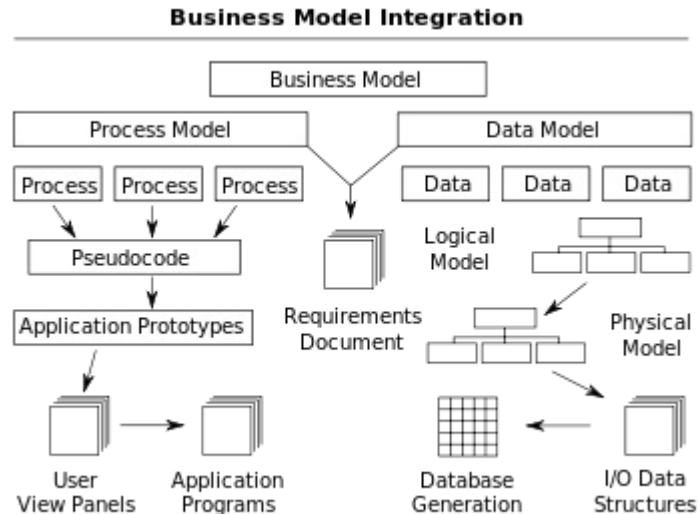
Factors Affecting Documentation



- ◆ Different audiences
 - Business Requirements
 - User Requirements
 - System Requirements
- ◆ Type of SDLC
 - Agile
 - Traditional
- ◆ Size of project
 - Small
 - Large
- ◆ Type of project
 - Regulatory
 - Tactical

Requirements Documentation Techniques

- ◆ Acceptance Criteria
- ◆ Data Flow Diagrams
- ◆ Functional Requirements
- ◆ Key Performance Indicators
- ◆ Process Flow Diagrams
- ◆ Non Functional Requirements
- ◆ Text-based Requirements
- ◆ Use Case Narratives
- ◆ Use Case Diagrams
- ◆ User Stories



Requirements Document Templates

- ◆ There is no one right way to document requirements
- ◆ Templates help
 - Industry standard templates
 - Company specific templates

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ISO/IEC/IEEE 29148

Chapter Summary

In this chapter, we have prepared requirements for allocation to a release by:

- Prioritizing requirements using a sensible methodology
- Organizing requirements to create logical groupings
- Identifying ways to document requirements
- Comparing and contrasting how requirements are prioritized in Agile and Waterfall